

Precalc Lesson 1.8 Homework

Answer Key

1. 11 store clerks are needed to attend to the shoppers 3 hours after the mall opens.
2. $O(f(t)) = \left(\frac{9}{5}t + 32\right) - 15$, where t is the temperature in $^{\circ}C$, $f(t)$ is the corresponding temperature in $^{\circ}F$ and $O(f(t))$ is the temperature at which this particular oven needs to be set when the cookbook calls for a temperature of t .
3. 3
4.
 - a. 0
 - b. $f(g(x)) = (-3x + 7)^2 - 1$
 - c. -2
 - d. $g(f(x)) = -3(x^2 - 1) + 7$
5. Many answers are possible. One possible solution is $h(x) = \frac{3}{x}, f(x) = x^2 - 7$.
6. A
7.
 - a. $f(x) = x - 5$
 - b. $g(x) = 0.8x$
 - c. $f(g(75)) = \$55$. This is the cost of your purchase if you apply coupon 2 first, and then coupon 1.
 - d. $g(f(75)) = \$56$. This is the cost of your purchase if you apply coupon 1 first, and then coupon 2.
 - e. I would want coupon 2 applied first because I want 20% taken off of the larger amount. If you take 20% off of a smaller amount, you don't save as much.
8. C
9.
 - a. 2
 - b. -12